Principles of Computer Architecture

CSC 461/661/761

Term: Fall 2025

Time: Monday & Wednesday, 12:30-1:45pm

Location: Petty 224

Credit Credits: 3 credits

Prerequisites: Grades of at least C (2.0) in CSC 261, CSC 330, and CSC 350, or permission of instructor.

Instructor: Richard Johnson

Office: Petty 006

Office Hours: Monday & Wednesday, 4:15pm-5:15pm

Email: rbjohns4@uncg.edu

Course Description

Hardware and software components of computer systems, their organization and operations. Topics: comparative instruction set architectures, microprogramming, memory management, processor management, I/O, interrupts, and emulation of processors.

Required Textbook: Computer Architecture: A Quantitative Approach, Hennessy and Patterson, 6th Edition, Morgan Kaufmann, (*5th Edition will work too*)

Optional reading: Computer Organization and Design, Patterson & Hennessy, 6th edition, Morgan Kaufmann, 2021

Other readings: provided in class or through canvas system.

Student Learning Outcomes

Upon successful completion of this course students will be able to

- Understand computer instruction set architecture
- Explain how arithmetic computations are performed inside CPU
- Solid understanding of computer processor's functionalities and hardware support
- Understand computer memory hierarchy in detail
- Differentiate the characteristics of various computer storage
- Understand how I/O devices and systems work
- Explain modern multi-core multi-processor systems.
- Work on some simple logic design

Course Topics (Subject to Change)

The following topics will be covered in this course (subject to change).

- Introduction to computer architecture
- Instruction set architecture
- Pipelining
- Memory hierarchy
- Instruction-level parallelism (Loop unrolling, Branch prediction, Tomasulo)
- Thread-level parallelism
- Data-level parallelism

Tentative Grading Policy

Your grade is based on the following criteria.

Category	CSC 461	CSC 661/761
Exam 1	20%	20%
Exam 2	20%	20%
Final Exam	25%	25%
Assignments	35%	
Technical Report		35%

Course Delivery

The format for this course is two 75-minute lectures per week, punctuated by occasional discussion, exercises, group work, and student presentations *(if we have time)*. Evaluation will be based on assignments, two midterm and one final exam, and *(for graduate students)* a technical report.

Assignments

Students are responsible for completing assigned readings, assignments, and preparing for exams outside of class. Assignments will be a mix of written analysis and applied programming. Written assignments should be turned in electronically. Handwritten submissions are strongly discouraged (if you feel typing equations is too tough, try to learn LaTeX or write on white papers and then submit a high-resolution snapshot of them before you insert them into your Word file).

- Prepare your submission as a single Word document.
- The use of Dropbox, Google Drive, or any similar methods, and sharing your document is not allowed.
- Please submit your files through Canvas' assignment. No comments should be added to the submission page because assignments will be downloaded to grade offline. Any thoughts / explanations should be added into the Word file that you are submitting.
- Keep in mind that the instructor interface and student interface on Canvas could be different. Email questions to me if you can't find the submission link or talk to your classmates.
- Avoid using unnecessary space or non-alphabetic characters in your filename. Suggested filenames are "Lastname1.docx", "Lastname2.docx", etc.
- If you need to submit multiple files, please put all files including your word file into a single folder, named "Lastname1" etc., and zip entire folder into a ".zip" file and submit the zip file, i.e., Lastname1.zip

For undergraduate students, I plan to give five to seven assignments to be graded. You will receive assignments and submit your completed work through Canvas as attached files, generally either PDFs or source code files as appropriate. Answers should be typed when feasible, but figures and complex formulas may be hand-drawn and scanned or photographed.

Graduate students will, typically in pairs, read primary literature related to the course and deliver brief reports.

Except for explicit team assignments, each student must turn in their own work. You may not copy other students' work or allow your own work to be copied. However, short of copying, you are welcome to collaborate with other students. Collaboration can mean giving or receiving help or working together toward ideas and solutions. You must specify in your work which other students you collaborated with.

You are also welcome to find outside resources, such as books, websites, or YouTube videos, and to share these resources with other students through online discussions. If you directly use information or ideas from outside study material in completing an assignment, you must cite that material in your work, as you would in any academic work. However, you may not use materials that give answers to specific homework questions or programming assignments.

Exams

No collaboration or communication is allowed during exams. All work must be your own. Any violations of this policy are violations of academic integrity, and the university's academic integrity policy, linked in a later section, applies.

Final Exam: Dec 8, 12-3PM, Petty 213

Final Exam Schedule:
https://reg.uncg.edu/wp-content/uploads/2025/06/FA-25-Exam-Schedule.pdf

Makeup Policy

You may not make up exams without prior arrangements, a written medical excuse or a documented emergency. Makeup exams are at the sole discretion of the instructor, if you fail to make up for the exam within a reasonable period of time, you will receive a zero.

Late submissions of homework are strongly discouraged, unless advanced permission is granted. There will be a late penalty of 15 percent if you submit it within 24 hours from the on-time due date. Under certain circumstances an extension may be granted if a homework assignment has become too difficult or there is a major time-conflict with other academic activities.

Academic Integrity Policy

Students are expected to be familiar with and abide by the UNCG Academic Integrity Policy, which is online at http://academicintegrity.uncg.edu/. Assignments in this class are for individual work, unless explicitly stated otherwise. General concepts and material covered in the class may be discussed with other students or in study groups, but specific assignments should not be discussed, and any submitted work should be entirely your own. It is expected that the textbook will be used as a reference, but if any other reference materials, especially those from the Internet, are used in preparing homework solutions they should be clearly cited. Any incidents of academic dishonesty will be handled strictly, resulting in either a zero on the assignment or an "F" in the class, depending on the severity of the incident, and incidents will be reported to the appropriate UNCG office. By submitting your homework or exam answer sheet, you have implicitly signed the Academic Integrity Pledge, stating that "I have abided by the UNCG Academic Integrity Policy on this assignment or submission."

If it is found that your homework or program is the result of plagiary from the Internet or your classmate's work, you will be reported (so will be the classmate whose work has been plagiarized).

Students are expected to abide by the UNCG's Academic Integrity Policy and the Student Code of Conduct:

- Student Conduct: http://sa.uncg.edu/handbook/student-code-of-conduct/
- Academic Integrity Policy: http://sa.uncg.edu/handbook/academic-integrity-policy/

Attendance Policy

Students are expected to attend class regularly. It is the student's responsibility to obtain the information that was presented during a class that was missed. Students should inform the instructor as far in advance as possible when conflicts or absences can be anticipated. (e.g., athletic team events, work, medical procedures, religious observances, etc.).

Canvas Policy

You should check Canvas regularly for course announcements. Grades from assignments and exams will be posted to Canvas. It is the student's responsibility to inform the instructor of missing or misreported grades within one week after they are posted. All course materials will be accessible through Canvas.

Email Communication

I may use email to communicate with you. I will only use your UNCG email addresses. Please make sure to check your messages regularly. If you normally use a different email address, it is your responsibility to forward your UNCG messages to that address. When you email me (or reply), please use only your UNCG email.

Laptop / Cellphone Policy:

Laptops can be both a benefit and a distraction in the classroom. While many students benefit from taking notes using a laptop or having access to outside class-related resources during class, other students cannot resist the temptation of checking e-mail, social texting, or even playing games during class time. This class has a strict "no non-class related use" rule for laptops or cell phones. Cell phones are a distraction for everyone and should be turned off or muted / vibrated during class.

E-cigarette:

Please do not smoke e-cigarettes in class.

Recording Policy

No student may record any classroom activity without express written consent from the instructor. If you have (or think you may have) a disability such that you need to record or tape classroom activities, you should contact the Office of Accessibility Resources & Services http://ods.uncg.edu/ to request appropriate accommodation.

Classroom Decorum

You should arrive on time; habitual tardiness is disruptive. Eating and drinking is not permitted in class. Smoking is strictly prohibited in the classroom. This includes e-cigarettes. Disruptive behavior will not be tolerated, you will be asked to leave. A pattern of disruptive behavior may result in your withdrawal from the course.

Withdrawal

A syllabus constitutes an agreement between the student and the course instructor about course requirements. Participation in this course indicates your acceptance of its teaching focus, requirements, and policies. Please review the syllabus and the course requirements as soon as possible. If you believe that the nature of this course does not meet your interests, needs or expectations, if you are not prepared for the amount of work involved or if you anticipate that the class meetings, assignment deadlines or abiding by the course policies will constitute an unacceptable hardship for you, you should drop the class by the drop/add deadline, which is located on the Academic Calendar. For more information, please visit the Office of the University Registrar.

ADA Statement

UNCG seeks to comply fully with the Americans with Disabilities Act (ADA). Students requesting accommodation based on a disability must be registered with the Office of Accessibility Resources and Services located in 215 Elliott University Center: (336) 334-5440 (or http://oars.uncg.edu).

University Closings

If university facilities are closed due to flu outbreak or other emergencies, it does not mean that classes are canceled. In such an event, please check the Canvas site for information about if and how the class will proceed.

Disclaimer

This syllabus is intended to give the student guidance on what may be covered during the semester and will be followed as closely as possible. However, the instructor reserves the right to modify, supplement and make changes as needs arise.